MD IRSAD

AI Engineer | Python Backend | GenAI & Automation Engineer

+91-8409718735 | mdirsadtech7305@gmail.com | LinkedIn | GitHub

PROFESSIONAL SUMMARY

AI-focused Python Engineer with 3+ years of experience building backend APIs, AWS data pipelines, workflow automation, finance-data applications, and practical LLM-enabled solutions. Hands-on with Python, FastAPI, Flask, REST APIs, SQL, Pandas, NumPy, PySpark, AWS Lambda, Glue, S3, Step Functions, SQS, DynamoDB, Redshift, CloudWatch, Docker, CI/CD, and automated testing. Experienced with LLM API integration, prompt-driven workflows, structured API automation, RAG concepts, data preparation, production monitoring, failure handling, and cloud-native application integration. Strongest fit is AI Engineering focused on Python backend, GenAI application development, AI automation, APIs, data pipelines, and reliable deployment rather than research-heavy deep-learning roles.

Core Expertise: Python | GenAI Application Engineering | LLM API Integration | Prompt Engineering | RAG Concepts | AI Automation | FastAPI | REST APIs | SQL | Pandas | NumPy | AWS | PySpark | ETL/ELT | Docker | CI/CD | Observability

TECHNICAL SKILLS

Programming & Backend: Python 3.12/3.13 | FastAPI | Flask | REST APIs | SQLAlchemy | Pydantic | Boto3 | SQL

AI & GenAI: LLM APIs | Prompt Engineering & Automation | RAG Concepts | Structured Outputs | AI-assisted Workflows | Data Preparation for ML | Context-aware API Workflows | Python Automation

Data Engineering & Analytics: Pandas | NumPy | PySpark | AWS Glue | ETL/ELT | Data Transformation | Schema Mapping | Incremental Processing | Data Validation | Data Pre-processing

AWS & Cloud: Lambda | S3 | Glue | Step Functions | SQS/SNS | DynamoDB | Redshift | Athena | CloudWatch | IAM | API Gateway | ECR | ECS | Cognito

Databases & DevOps: MySQL | PostgreSQL | DynamoDB | Redshift | Docker | Git | GitHub | Poetry | CI/CD | Pytest | Unittest | Testcontainers | SonarQube

Frontend & Tools: React.js | Next.js | Tailwind CSS | JavaScript/ES6+ | Axios | Jupyter Notebook | Postman | DBeaver | VS Code | PyCharm

PROFESSIONAL EXPERIENCE

Data Engineer | Freelance - Client: HCLTech / MOTOR

Feb 2025 - Present

Cloud Data, Python APIs & Automation

Build Python and AWS-based data workflows, API integrations, document automation, serverless processing, monitoring, and production issue handling across finance-data projects, with transferable patterns for AI-enabled cloud applications.

Key Contributions:

- Developed AWS Glue/PySpark ETL pipelines for invoice, remittance, and credit-memo processing with schema mapping, validation, transformation, and incremental loads.
- Orchestrated serverless workflows using Step Functions, Lambda, S3, SQS, DynamoDB, Redshift, and REST APIs, applying retry, error-handling, and monitoring patterns.
- Integrated Oracle ORDS/REST APIs and automated document/data workflows using Python, including structured payload processing and downstream service integration.
- Used CloudWatch logging and alerts to troubleshoot failures, validate data quality, and improve reliability of cloud workflows.

Tech Stack: Python | PySpark | Pandas | AWS Glue | Lambda | S3 | SQS | Step Functions | DynamoDB | Redshift | CloudWatch | Oracle ORDS | REST APIs

Software Developer | 33 Kuber India Advisory LLP

Sep 2023 - Jan 2025

Python Backend, APIs & Business Applications

Developed Python backend services and full-stack applications with REST APIs, MySQL, React, authentication, reporting, workflow automation, testing, and database optimization.

Key Contributions:

- Built and maintained Python REST APIs using Flask/FastAPI patterns, MySQL, validation, authentication, and reusable backend services.
- Developed OTP login, password reset, CRM/CRS/SRC integrations, task, attendance, timesheet, and leave workflows across business applications.
- Optimized SQL and MySQL using stored procedures, views, functions, indexing, and query improvements for reliable application data access.
- Integrated React/Tailwind frontends with backend APIs and supported automated testing, debugging, and production issue resolution.

Tech Stack: Python | Flask | FastAPI | MySQL | REST APIs | React.js | Tailwind CSS | SQL | Git | Pytest

PROJECTS

ALGOAGENTX - Python FinTech Automation Platform

Role: Independent Developer | **Focus:** Backend APIs, automation, analytics, execution workflows

Built a Python/FastAPI platform for backtesting, market-data workflows, broker connectivity, trading-rule automation, risk controls, reporting, and operational dashboards, demonstrating end-to-end API and automation engineering applicable to AI-enabled products.

- Implemented FastAPI backend services, PostgreSQL/Redis workflows, backtest execution, trade/PnL analytics, funded-account rule checks, and report generation.
- Integrated broker connectivity through an MT5 agent, symbol mapping, queue/worker processing, and event-driven alert workflows.
- Added configurable risk controls, execution safeguards, monitoring, and operational status handling for long-running automation jobs.
- Built Next.js dashboards for backtest history, reports, live workflows, strategy administration, and application operations.
- Used Dockerized services and API-driven architecture to separate backend, frontend, Redis workers, database, and integration components.

Tech Stack: Python | FastAPI | PostgreSQL | Redis | Docker | REST APIs | Next.js | WebSocket | Automation | Analytics

PRINTSOFT REPLACEMENT - Serverless Finance Document Automation

Client: AUS Logistics | **Period:** May 2025 - Present | **Role:** Data Engineer

Migrated a legacy finance-document workflow to serverless AWS processing with automated API ingestion, enrichment, document generation, storage, and monitoring.

- Built Lambda/SQS/S3 workflows for invoices, bills, statements, and credit memos with Oracle ORDS API integration.
- Implemented DynamoDB enrichment, conditional PDF/CSV templates, naming rules, reprint logic, and audit-friendly output storage.
- Applied serverless monitoring and error-handling patterns transferable to production AI and API workloads.

Tech Stack: Python | AWS Lambda | SQS | S3 | DynamoDB | Oracle ORDS | REST APIs | PDF/CSV Automation

PROD FINANCE DATA PIPELINE - AWS Data Engineering

Client: MOTOR | **Period:** Feb 2025 - May 2025 | **Role:** Data Engineer

Designed and deployed an AWS data pipeline for finance datasets using distributed ETL, orchestration, cloud storage, monitoring, and data-quality controls.

- Developed Glue/PySpark ETL for approximately 30GB+ daily data with schema mapping, incremental loads, bookmarks, and validation.
- Orchestrated processing with Step Functions and Lambda, using S3 as the data-lake layer and CloudWatch for operational monitoring.

Tech Stack: Python | PySpark | AWS Glue | S3 | Lambda | Step Functions | DynamoDB | Redshift | CloudWatch

AI / AUTOMATION PROFILE

Applied AI Focus: Practical experience with LLM APIs, prompt automation, RAG concepts, structured backend workflows, data preparation, and Python-based automation. Comfortable integrating AI services into APIs and cloud workflows; currently strongest in application engineering and orchestration rather than model training or deep-learning research.

Relevant Transferable Skills: Python | FastAPI | REST APIs | LLM APIs | Prompt Engineering | RAG Concepts | Structured Outputs | SQL | Pandas | NumPy | AWS | Docker | CI/CD | CloudWatch | Testing | Workflow Automation

EDUCATION

B.Sc. Information Technology — University of Mumbai | Sree Narayana Guru College of Commerce | Graduated 2023 | 9.13 CGPA

Class XII — R.R.M.Y Inter College High School+2 | Board of Intermediate Education | 2020 | 73.2%

Class X — S.R.P.N. High School | Board of Secondary Education | 2018 | 63%

TARGET ROLES

Target Roles: AI Engineer | GenAI Engineer | AI Automation Engineer | Python AI Backend Engineer | Applied AI Engineer | Backend Engineer | Data Engineer